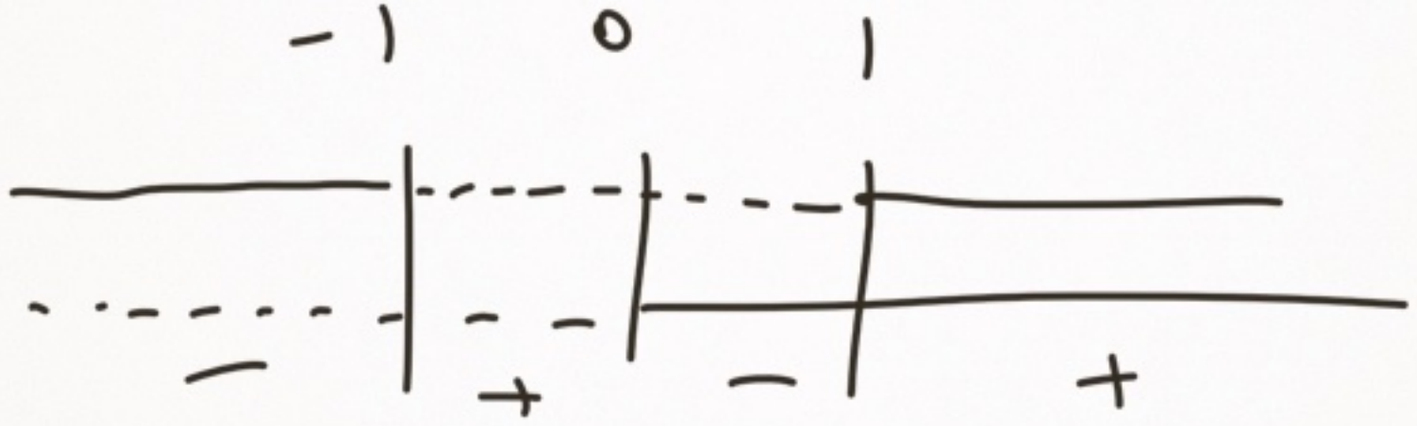
(1) $f(x) = \ln\left(\frac{x^2-1}{x}\right)$

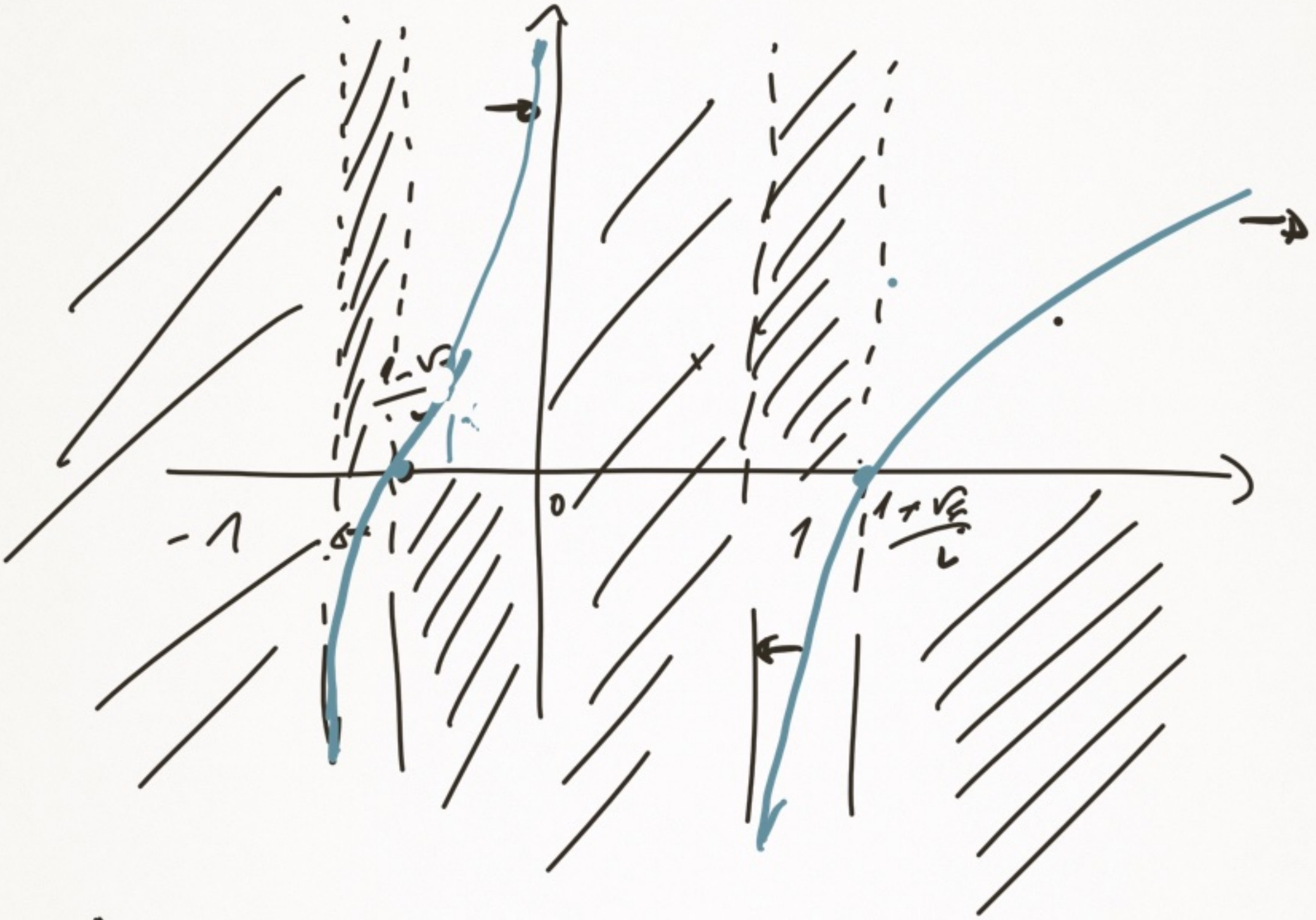
Dominio: $\frac{x^2-1}{x} > 0$

$x^2 - 1 > 0 \Rightarrow x < -1 \vee x > 1$

$x > 0$

$-1 < x < 0 \vee x > 1$





Dominio non simmetrico
f non è né pari né dispari.

$x = 0$ non appartient au domaine, non ci sono limitazioni per l'ora

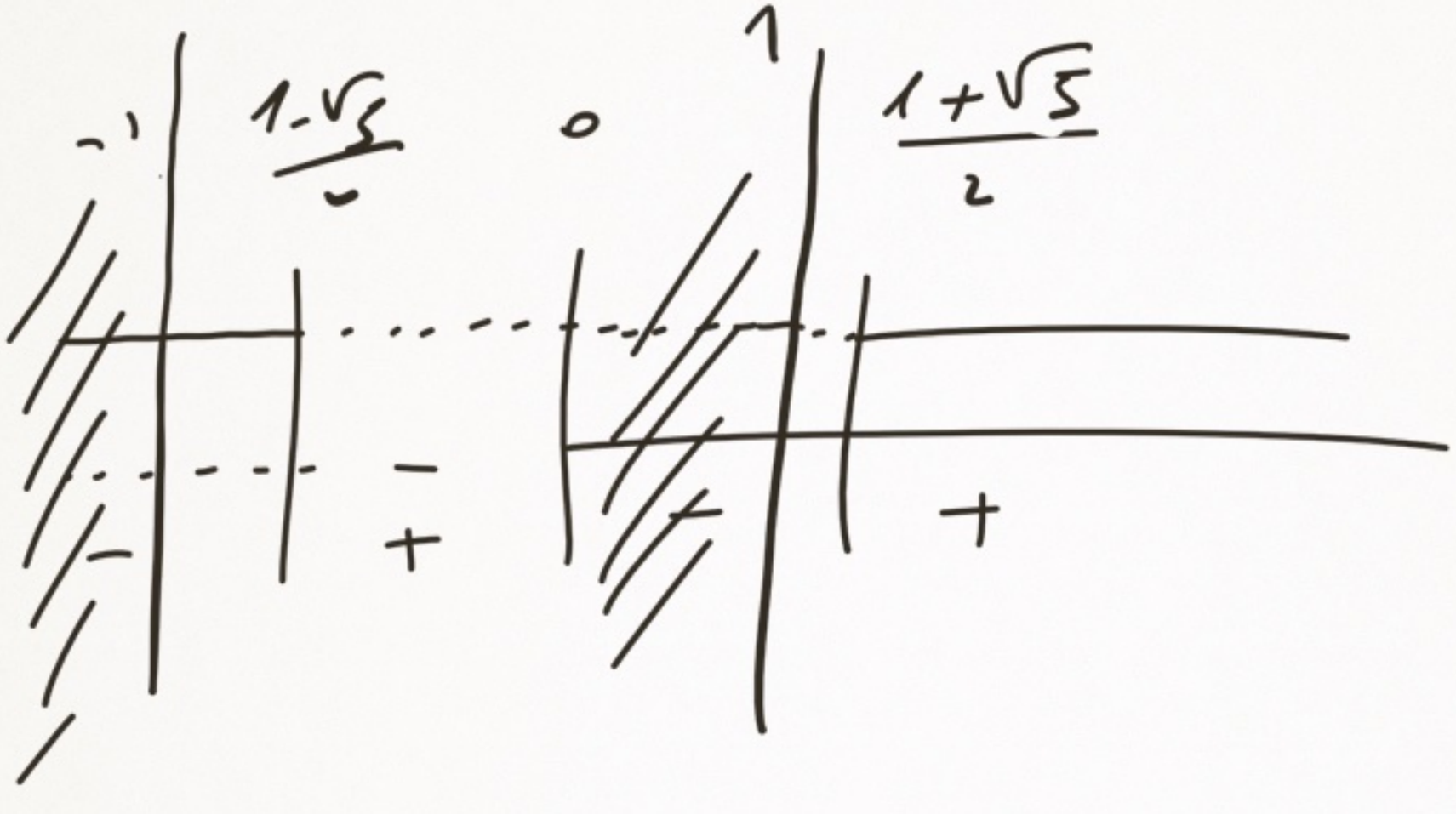
- $f(x) \geq 0$

$$\ln\left(\frac{x-1}{x}\right)^2 \geq 0$$

$$\frac{x-1}{x} \geq 1 \iff \frac{x-x-1}{x} \geq 0$$

- $x^2 - x - 1 \geq 0 \iff x \leq \frac{1-\sqrt{5}}{2} \vee x \geq \frac{1+\sqrt{5}}{2}$

- $x > 0$



$f(x) \geq 0 \iff \frac{1-\sqrt{5}}{2} \leq x < 0$

$\vee \quad x > \frac{1+\sqrt{5}}{2}$

$$\lim_{x\to -1^+} f(x) = \lim_{x\to -1} \ln\left(\frac{x^2-1}{x}\right) = -\infty$$

$$\lim_{x\to -1^+} \frac{x^2-1}{x} = 0^+$$

$x = -1$ ASINTOTO VERTICALE DESTRO

$$\lim_{x\to 0^-} \ln\left(\frac{x^2-1}{x}\right) = \ln\left(\frac{-1}{0^-}\right)$$

$$= \ln(+\infty) = +\infty$$

$x = 0$ ASINTOTO VERT. SINISTRO

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \ln\left(\frac{0^+}{1}\right) = -\infty$$

$x = 1$ ASINTOTO VERT. DESTRO

$$\lim_{x\to +\infty} \ln\left(\frac{x^2-1}{x}\right) = \lim_{x\to +\infty} \ln\left(\frac{x^2}{x}\right) =$$

$$= \lim_{x\to +\infty} \ln x = +\infty$$

NO ASINTOTI ORIZZONTALI,

CERCO L'EVENTUALE OBLIQUO

$$m = \lim_{x\to+\infty} \frac{f(x)}{x} =$$

$$= \lim_{x\to+\infty} \frac{\ln\left(\frac{x^2-1}{x}\right)}{x} =$$

$$= \lim_{x\to+\infty} \frac{\ln x}{x} = 0$$

Non esiste asintoto obliquo.

Passiamo a studiare la monotonia

$$f(x) = \ln\left(\frac{x^2-1}{x}\right)$$

$$f'(x) = \frac{1}{\frac{x^2-1}{x}} \cdot \frac{2x^2 - (x^2-1)}{x^2} =$$

$$= \frac{x}{x^2-1} \cdot \frac{2x^2 - x^2 + 1}{x^2} = \frac{x^2+1}{x(x^2-1)}$$

Dominio: $x \neq 0,\ x \neq \pm 1$,

f è derivabile nel dominio

$$f'(x) \geq 0$$

$$\frac{x^2+1}{x^2(x-1)} \geq 0$$

$x^2+1 > 0 \quad \forall x \in \mathbb{R}$

$x^2(x-1) > 0 \iff \frac{x-1}{x^2} > 0 \iff$

$\iff -1 < x < 0 \ \vee \ x > 1$

Dominio di f

Allora $f' > 0$ nel dominio $\Rightarrow f$ strett. crescente

$$f'(x) = \frac{x^2+1}{x(x^2-1)} = \frac{x^2+1}{x^3-x}$$

$$f''(x) = \frac{2x(x^3-x) - (x^2+1)(3x^2-1)}{(x^3-x)^2} =$$

$$= \frac{2x^4 - 2x^2 - (3x^4 - x^2 + 3x^2 - 1)}{(x^3-x)^2} =$$

$$= \frac{2x^4 - 2x^2 - 3x^4 - 2x^2 + 1}{(x^3-x)^2} = \frac{-x^4 - 4x^2 + 1}{(x^3-x)^2}$$

$$f''(x) \geq 0$$

$$-\frac{x^4 - 4x^2 + 1}{(x^3 - x)^2} \geq 0 \quad (=) \quad -x^4 - 4x^2 - 1 \geq 0$$

$(x^3 - x)^2$: >0

$$(=) \quad x^4 + 4x^2 - 1 \leq 0$$

$$t = x^2$$

$$t^2 + 4t - 1 \leq 0$$

$$t_{1,2} = \frac{-4 \pm \sqrt{16 + 4}}{2} = \frac{-4 \pm \sqrt{20}}{2}$$

$$t^2+4t-1 \leq 0 \iff$$

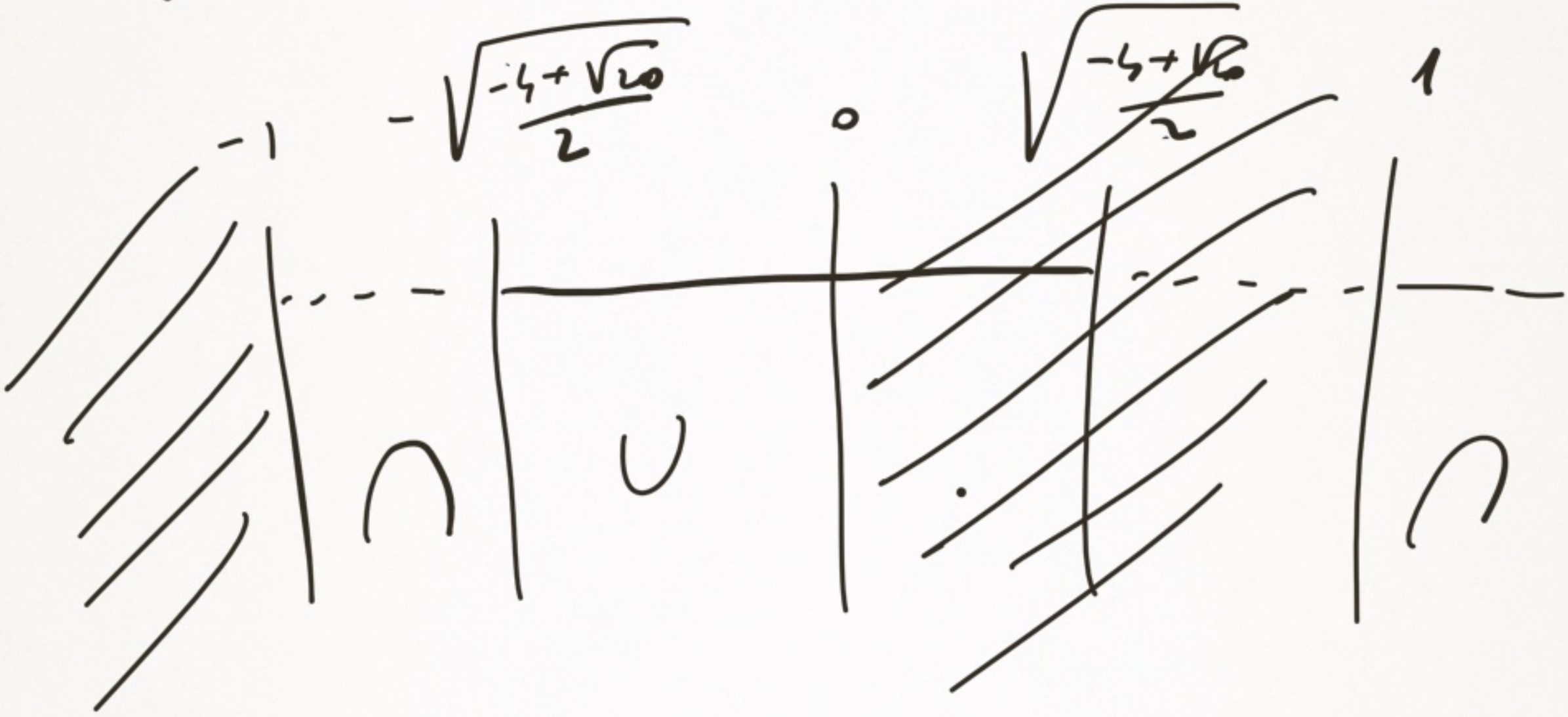
$$\iff \underbrace{\frac{-4-\sqrt{20}}{2}}_{<0} \leq t \leq \underbrace{\frac{-4+\sqrt{20}}{2}}_{>0}$$

$t = x^2$, quoi'

$$\frac{-4-\sqrt{20}}{2} \leq x^2 \leq \frac{-4+\sqrt{20}}{2}$$

$$\begin{cases} x^2 \geq \frac{-4-\sqrt{20}}{2} \rightarrow \forall x \in \mathbb{R} \\ x^2 \leq \frac{-4+\sqrt{20}}{2} \iff -\sqrt{\frac{-4+\sqrt{20}}{2}} \leq x \leq \sqrt{\frac{-4+\sqrt{20}}{2}} \end{cases}$$

$f'' \geq 0 \Rightarrow$



$$\int f'(x)\,g(x)\,dx = f(x)\cdot g(x) - \int f(x)\cdot g'(x)\,dx$$

$$g(x) = \cos x \Rightarrow g'(x) = -\sin x$$

$$f'(x) = x \Rightarrow f(x) = \frac{x^2}{2}$$

(2) $\int x^2 \cos x\,dx$

5. Intégrer chacune de façons suivantes :

$$\int_{x_1}^{x_2} \sin x \, dx$$

$$g(x) = x \quad \Rightarrow \quad g'(x) = 1$$

$$f(x) = \sin x \quad \Rightarrow \quad f'(x) = \cos x$$

$$= \frac{x_2}{2} \cos x \Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} \frac{1}{2} \sin x \, dx$$

$$\int_{x_1}^{x_2} \cos x \, dx = \frac{x}{2} \cdot \cos x \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{x}{2} (-\sin x) \, dx$$

$$\int_{x_1}^{x_2} \sin x \, dx = \frac{\lambda}{2} \int_{x_1}^{x_2} \cos x \, dx - \frac{\lambda}{2} \int_{x_1}^{x_2} \cos x \, dx =$$

$$= \left(\frac{\lambda}{2} \sin x \right)_{x_1}^{x_2} + \frac{\lambda}{2} \cos(\omega) \int_{x_1}^{x_2} = \int_{x_1}^{x_2} x_c \cos(\omega x) \, dx$$

Allora

$$= \frac{1}{5}\left(\frac{2}{2}\cos 2x + \sin\frac{x}{2}\right)\Big|_{x_1}^{x_2} + c$$

$$\int_{x_1}^{x_2} \frac{\sin 3x}{\cos^2 x}\,dx = \int_{x_1}^{x_2} 2\left(\frac{2}{2}\cos 2x + \frac{1}{2}\sin\frac{x}{2}\right)\Big|_{x_1}^{x_2} + c$$

$$= \frac{2}{2}\cos 2x + \frac{2}{2}\sin\frac{x}{2}$$

(3) Studiare la funzione

$$f(x) = \begin{cases} x^2 + 2x & 0 \leq x \leq 1 \\ e^{2x} - 1 & -1 \leq x < 0 \end{cases}$$

.

verifica le ipotesi del teorema di Lagrange, ossia

(1) f continua in $[-1, 1]$

(2) f derivabile in $]-1, 1[$

Dobbiamo studiare la continuità e la derivabilità in $x=0$, perché per $x \neq 0$ f è continua ed è derivabile f è continua in 0 $\Leftrightarrow$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = f(0)$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2 + 2x = 0 \qquad f(0) = 0$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} e^{2x} - 1 = 0$$

f è continua in $x = 0$

Usiamo il criterio del confronto

$$f'(x) = \begin{cases} 2x + 2 & 0 < x \leq 1 \\ 2x & \\ 2e & -1 < x < 0 \end{cases}$$

$$\lim_{x \to 0^-} f'(x) = \lim_{x \to 0^-} 2e = 2e = 2$$

$$\lim_{x \to 0^+} f'(x) = \lim_{x \to 0^+} 2x + 2 = 0 + 2 = 2$$

Allora f è derivabile in 0 e $f'(0) = 2$

Studiare la convergenza di

$$\sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^4}} \sin^2\left(\frac{2}{n^2}\right)$$

Serie a termini positivi.

$$\lim_{n\to+\infty} \frac{n+1}{\sqrt{n^4}} \cdot \sin^2\left(\frac{2}{n^2}\right) =$$

$$= \lim_{n\to+\infty} \frac{n^2}{n^2} \left(\frac{2}{n^2}\right)^2 =$$

$$= \lim_{n\to+\infty} \frac{4}{n^4}$$

$$\sin\left(\frac{2}{n^2}\right) \sim \frac{2}{n^2}$$

La serie è equivalente a

$$\sum_{n=1}^{\infty} \frac{4}{n^4} = 4 \underbrace{\sum_{n=1}^{\infty} \frac{1}{n^4}}$$

$\downarrow$

SERIE ARMONICA GENERALIZZATA con $\alpha = 4 > 1$

Allora la serie converge.

(5) STABILIRE SE LA FUNZIONE

$$f(x) = x^2 + x \quad , \qquad x \in [0,1]$$

SODDISFA LE IPOTESI DEL

TEOREMA DEI VALORI INTERMEDI,

E SI CALCOLI $f([0,1])$

— ~

TEOR: $f:[a,b] \to \mathbb{R}$ continua.

Allora $f([0,1]) = [m, M]$

$m = \min_{x \in [a,b]} f(x)$, $M = \max_{x \in [a,b]} f(x)$

f è definita in $[0,1]$

f è continua perché è polinomio.

Allora f verifica le ipotesi del

teorema dei valori intermedi.

Devo calcolare m, M

$$f(x) = x^2 + x$$

$$f'(x) = 2x + 1 \geq 0 \iff x \geq -\frac{1}{2}$$



$f' > 0 \quad \forall x \in [0,1]$

$\Rightarrow$ f è strett. monotona in crescita in $[0,1]$

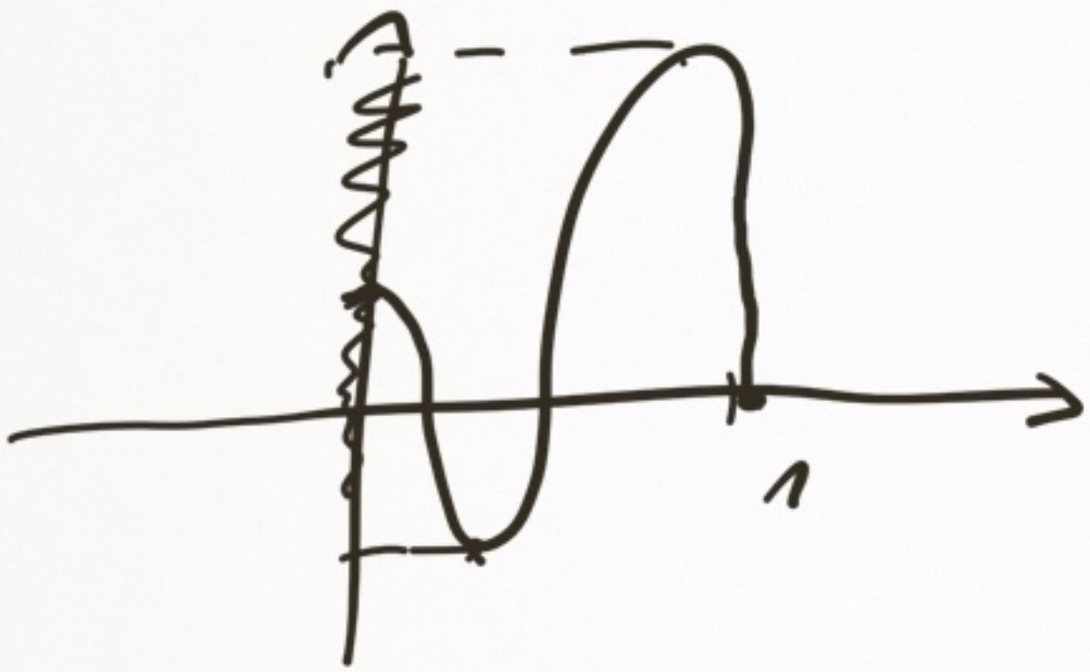
$$\min_{x \in [0,1]} f(x) = f(0) = 0$$



$$\max_{x \in [0,1]} f(x) = f(1) = 1$$

$$f([0,1]) = [m, M] = [0,1]$$

OSS VALE SOLO PER f
CONTINUE CRESCENTE



con $f = [f(0), f(1)]$

Vale per f STRETT. CRESCENTE